

The Hubbard dimer from partial spectral functions

The Hubbard atom calculation of

`hubbard_atom_spectral_representation.ipynb`, repeated for two sites.

The spectral machinery of

F. B. Kugler, S.-S. B. Lee and J. von Delft, *Multipoint Correlation Functions: Spectral Representation and Numerical Evaluation*, [Phys. Rev. X **11**, 041006 \(2021\)](#)

is reused unchanged: PSFs from Eq. (28), the Matsubara kernel Eq. (46), the Keldysh kernels Eqs. (52), (67b) and assembly Eq. (67a). Only the Hamiltonian, the operators and the momentum bookkeeping of the vertex are new.

What the dimer tests that the atom could not.

1. *Nonlocal operators.* Eq. (28) is a Lehmann sum over arbitrary operators $\mathcal{O}^i$; locality enters the paper only from Sec. III on. Here the operators are site and momentum combinations.
2. *Momentum.* A selection rule, $k_1 - k_2 + k_3 - k_4 \equiv 0 \pmod{2\pi}$, that a single site cannot have.
3. *A nondegenerate singlet ground state* instead of the atom's free-moment doublet. That changes what the $\beta\delta_\omega$ terms of Eq. (85) do (D11).

Provenance. No paper in the project treats the dimer. The closed forms used as benchmarks (spectrum, Z , G_k) come from the task sheet; this notebook checks them against a 16-state exact diagonalisation rather than assuming them. Every equation cited from the paper was read from the rendered page image, not the PDF text layer.

Check	Content	Result
D1	spectrum and Z	$\leq 5\text{e-}15$
D2	$G_k(i\nu)$ from the $\ell = 2$ PSFs vs the closed form	$\leq 4\text{e-}15$
D3	G_{11} , G_{12} from site operators	$\leq 1\text{e-}15$; G_{11} purely imaginary
D4	Keldysh $\ell = 2$: G^R , G^A ; FDT (C4) at every pole	$\leq 1\text{e-}13$ relative; $O(\gamma_0^2)$
D5	momentum selection rule, 8 forbidden tuples	exactly 0; $\leq 8\text{e-}16$ unfiltered
D6	$U = 0$ null test	$\leq 3\text{e-}14$
D7	$F/U \rightarrow \frac{1}{2}\delta_{k_1-k_2+k_3-k_4}$	$\leq 6\text{e-}6$ at $U \leq 10^{-2}$

Check	Content	Result
D8	$t \rightarrow 0: F \rightarrow \frac{1}{2} F^{\text{atom}} \delta$; the two branches of Eq. (46)	exact at $t = 0$; $\propto t^2$ for $t \geq 10^{-3}$; roundoff $\propto 1/t^2$ below
D9	crossing symmetry and SU(2)	$\leq 5\text{e-}14$
D10	Keldysh: D5, D6, Eq. (69)	exact / $\leq 5\text{e-}14$ / exact
D11	size of the $\delta_{\omega_{12},0}$ term vs $\beta(c - U/2)$	cut off as $e^{-\beta(c-U/2)}$

The test suite (`julia --project test/runtests.jl`) runs the same checks as assertions. Every cell below runs in a few seconds; the whole notebook takes about a minute.

Running on Colab: the first cell includes `../src/HubbardAtom.jl`, which pulls in every file of `src/` (`psf.jl`, `kernel.jl`, `correlator.jl`, `keldysh.jl`, `dimer.jl`, `heatmap.jl`). Upload all seven files and point the `include` at them, or clone the repository; `src/` needs the `StaticArrays` package (`Pkg.add("StaticArrays")`).

In [1]:

```
include(joinpath(@__DIR__, "..", "src", "HubbardAtom.jl"))
using .HubbardAtom
using LinearAlgebra
using Printf
```

Model and operators

$$H = -t \sum_{\sigma} (c_{1\sigma}^{\dagger} c_{2\sigma} + c_{2\sigma}^{\dagger} c_{1\sigma}) + U \sum_{i=1,2} n_{i\uparrow} n_{i\downarrow} - \frac{U}{2} \sum_{i\sigma} n_{i\sigma},$$

$$c_{k\sigma} = \frac{1}{\sqrt{2}} (c_{1\sigma} + e^{ik} c_{2\sigma}), \quad k \in \{0, \pi\}.$$

The last term of H puts the chemical potential at half filling, as $\varepsilon_d = -U/2$ did for the atom. This is Eq. (80) of the paper, restricted to two sites, plus that shift. With hopping $-t$, $k = 0$ is the bonding orbital (energy $-t$) and $k = \pi$ the antibonding one ($+t$). Momenta are carried as an integer q with $k = q\pi$, so the selection rule is the exact test `iseven(q1+q2+q3+q4)`.

The four modes are ordered $1\uparrow, 1\downarrow, 2\uparrow, 2\downarrow$ and built with global Jordan-Wigner strings. The hopping term $c_{1\uparrow}^{\dagger} c_{2\uparrow}$ crosses mode $1\downarrow$ and carries its string sign. It gets that sign automatically because H is formed as a product of the mode matrices; no matrix element is entered by hand.

First, before anything else, the canonical anticommutators.

In [2]:

```
m = HubbardDimerModel(0.7, 2.3, 3.0)
ops = operators(m)
```

```

I16 = Matrix(1.0I, 16, 16)
car = maximum(max(norm(ops.c[a]*ops.c[b]' + ops.c[b]'*ops.c[a] - (a == b)*I16),
                 norm(ops.c[a]*ops.c[b] + ops.c[b]*ops.c[a])) for a in 1:4, b in 1:4)
@printf("{c_i, c_j†} = δ_ij and {c_i, c_j} = 0 for all four modes:   max deviation %.1e\n", car)

bad = jordan_wigner(4; strings = false) # the same construction without strings
@printf("without Jordan-Wigner strings, ||{c_1†, c_2†}|| = %.2f <- the assertion catches this\n\n",
        norm(bad[1]*bad[3] + bad[3]*bad[1]))

P = site_exchange(ops)
@printf("[H, N] = %.1e   [H, Sz] = %.1e   [H, P] = %.1e   (P = site exchange 1 ↔ 2)\n",
        norm(ops.H*ops.N - ops.N*ops.H), norm(ops.H*ops.Sz - ops.Sz*ops.H), norm(ops.H*P
        - P*ops.H))
@printf("P c_1σ P = c_2σ: %s   P c_0σ P = c_0σ: %s   P c_πσ P = -c_πσ: %s\n",
        P*site_op(ops,1,:up)*P ≈ site_op(ops,2,:up), P*momentum_op(ops,0,:up)*P ≈ momentu
        m_op(ops,0,:up),
        P*momentum_op(ops,1,:up)*P ≈ -momentum_op(ops,1,:up))

```

{c_i, c_j†} = δ_ij and {c_i, c_j} = 0 for all four modes: max deviation 0.0e+00
without Jordan-Wigner strings, ||{c_1†, c_2†}|| = 4.00 <- the assertion catches this

[H, N] = 0.0e+00 [H, Sz] = 0.0e+00 [H, P] = 1.6e-15 (P = site exchange 1 ↔ 2)
P c_1σ P = c_2σ: true P c_0σ P = c_0σ: true P c_πσ P = -c_πσ: true

D1 — spectrum and partition function

With $c \equiv \sqrt{U^2/4 + 4t^2}$ and $\alpha_{\pm} \equiv \frac{1}{2}(1 \pm 2t/c)$, the only nontrivial block is the pair of $N = 2$ spin singlets (covalent and symmetric-ionic), with eigenvalues $U/2 \pm c$ before the $-UN/2$ shift. The closed forms to reproduce:

$$Z = 3 + 3e^{\beta U} + 8e^{\beta U/2} \cosh \beta t + 2e^{\beta U/2} \cosh \beta c.$$

At large β , $e^{\beta U}$ and $\cosh \beta c$ overflow. The Boltzmann weights are therefore formed with energies measured from $E_0 = -U/2 - c$. Every exponent of $Ze^{\beta E_0}$ is then ≤ 0 , because $c \geq U/2$ and $c \geq 2|t|$.

In [3]:

```

for (t, U, β) in [(0.7, 2.3, 3.0), (0.3, 4.0, 10.0), (1.0, 0.5, 1.0)]
    m = HubbardDimerModel(t, U, β); sp = spectrum(m)
    @printf("(t, U, β) = (%.1f, %.1f, %.0f)\n", t, U, β)
    names = ["N=2 singlet (ground)", "N=2 triplet", "N=1,3 bonding", "N=1,3 antibonding",
            "N=0, N=4, ionic singlet", "N=2 excited singlet"]
    for ((E, d), name) in zip(dimer_levels_analytic(m), names)
        found = count(x -> abs(x - E) < 1e-9, sp.E)
        @printf("   %-24s E = %+8.5f   degeneracy %d, found %d\n", name, E, d, found)
    end
    lev = sort(vcat([fill(E, d) for (E, d) in dimer_levels_analytic(m)]...))
    @printf("   max |E - E_exact| = %.1e   |Z/Z_exact - 1| = %.1e\n\n",
            maximum(abs, sp.E - lev), abs(sp.Z / dimer_partition_function_analytic(m) -
1))
end
m = HubbardDimerModel(0.3, 4.0, 1000.0)
@printf("β = 1000: Z = %s, Z e^{βE_0} = %.6f, all p finite: %s\n",
        dimer_partition_function_analytic(m), dimer_partition_function_analytic(m; shifte
d = true),
        all(isfinite, spectrum(m).p))

```

(t, U, β) = (0.7, 2.3, 3)
N=2 singlet (ground) E = -2.96177 degeneracy 1, found 1

```

N=2 triplet           E = -2.30000    degeneracy 3, found 3
N=1,3 bonding         E = -1.85000    degeneracy 4, found 4
N=1,3 antibonding     E = -0.45000    degeneracy 4, found 4
N=0, N=4, ionic singlet E = +0.00000    degeneracy 3, found 3
N=2 excited singlet   E = +0.66177    degeneracy 1, found 1
max |E - E_exact| = 2.4e-15    |Z/Z_exact - 1| = 2.4e-15

(t, U, β) = (0.3, 4.0, 10)
N=2 singlet (ground)   E = -4.08806    degeneracy 1, found 1
N=2 triplet           E = -4.00000    degeneracy 3, found 3
N=1,3 bonding         E = -2.30000    degeneracy 4, found 4
N=1,3 antibonding     E = -1.70000    degeneracy 4, found 4
N=0, N=4, ionic singlet E = +0.00000    degeneracy 3, found 3
N=2 excited singlet   E = +0.08806    degeneracy 1, found 1
max |E - E_exact| = 4.4e-15    |Z/Z_exact - 1| = 3.1e-15

(t, U, β) = (1.0, 0.5, 1)
N=2 singlet (ground)   E = -2.26556    degeneracy 1, found 1
N=2 triplet           E = -0.50000    degeneracy 3, found 3
N=1,3 bonding         E = -1.25000    degeneracy 4, found 4
N=1,3 antibonding     E = +0.75000    degeneracy 4, found 4
N=0, N=4, ionic singlet E = +0.00000    degeneracy 3, found 3
N=2 excited singlet   E = +1.76556    degeneracy 1, found 1
max |E - E_exact| = 4.0e-15    |Z/Z_exact - 1| = 1.4e-15

β = 1000: Z = Inf, Z e^{βE_0} = 1.000000, all p finite: true

```

D2 — the propagator $G_k(i\nu)$

Four poles contribute to $G_{k=0}$:

$$G_{k=0}(i\nu) = \frac{1}{Z} \left\{ 2e^{\beta U/2} (\cosh \beta t + \cosh \beta c) \left[\frac{\alpha_+}{i\nu - (t - c)} + \frac{\alpha_-}{i\nu - (t + c)} \right] + \frac{3}{2} (1 + e^{\beta U} + 2e^{\beta U/2} \cosh \beta t) \left[\frac{1}{i\nu - (U/2 - t)} + \frac{1}{i\nu + (U/2 + t)} \right] \right\},$$

and particle-hole symmetry gives $G_{k=\pi}(i\nu) = -G_{k=0}(-i\nu)$. The computed side is Eq. (39) with $\mathcal{O} = (c_{k\sigma}, c_{k\sigma}^\dagger)$, the same $\ell = 2$ code as for the atom. The closed form is evaluated with the E_0 shift as well.

Trap 1. A wrong hopping sign leaves the spectrum and Z unchanged, because the spectrum is even in t . It exchanges G_0 and G_π , though, and D2 catches that.

In [4]:

```

for (t, U, β) in [(0.7, 2.3, 3.0), (0.3, 4.0, 10.0), (1.0, 0.5, 1.0)]
    m = HubbardDimerModel(t, U, β); ctx = DimerContext(m)
    e = maximum(abs(dimer_propagator(ctx, q, fermionic_freq(n); σ = σ) - dimer_propagator_
        _exact(m, q, fermionic_freq(n)))
        for q in 0:1, n in -6:5, σ in (:up, :dn))
    @printf("(t, U, β) = (%.1f, %.1f, %4.1f): max |G_k - closed form| over k ∈ {0,π}, n
        = -6...5, both spins = %.1e\n",
        t, U, β, e)
end

println("\nlimits of the closed form, each derivable by hand:")
m = HubbardDimerModel(0.8, 0.0, 2.0)
@printf("  U = 0:   G_0(0.3i) = %s, 1/(0.3i + t) = %s\n", round(dimer_propagator_exact
    (m, 0, 0.3im); digits=12), round(1/(0.3im + 0.8); digits=12))
m = HubbardDimerModel(0.0, 2.6, 4.0)
@printf("  t = 0:   G_0(0.7i) = %s, atom 1/2 Σ±(iν ± U/2)^-1 = %s\n", round(dimer_propagato
    r_exact(m, 0, 0.7im); digits=12),

```

```

        round(0.5*(1/(0.7im + 1.3) + 1/(0.7im - 1.3)); digits=12))
m = HubbardDimerModel(0.4, 2.0, 200.0); ap, am = dimer_alpha(m); c = dimer_c(m)
@printf("    T → 0:    G0(0.5i) = %s,    α/(iv-(t-c)) + α/(iv-(t+c)) = %s\n", round(dimer_
propagator_exact(m, 0, 0.5im); digits=12),
        round(ap/(0.5im - (0.4 - c)) + am/(0.5im - (0.4 + c)); digits=12))
m = HubbardDimerModel(0.7, 2.3, 3.0)
@printf("    sum rule: z G0(z) at z = 107i → %.8f\n", real(1e7im * dimer_propagator_exact
(m, 0, 1e7im)))

mw = HubbardDimerModel(-0.7, 2.3, 3.0); cw = DimerContext(mw); v = fermionic_freq(0)
println("\ntrap 1, hopping sign flipped (t → -t):")
@printf("    spectrum unchanged: %s\n", spectrum(mw).E ≈ spectrum(m).E)
@printf("    |G0(-t) - G0 closed form| = %.3f    <- D2 fails\n", abs(dimer_propagator(cw,
0, v) - dimer_propagator_exact(m, 0, v)))
@printf("    |G0(-t) - Gπ closed form| = %.1e    <- because G0 and Gπ were exchanged\n", a
bs(dimer_propagator(cw, 0, v) - dimer_propagator_exact(m, 1, v)))

```

(t, U, β) = (0.7, 2.3, 3.0): max |G_k - closed form| over k ∈ {0,π}, n = -6...5, both spins = 8.3e-16
(t, U, β) = (0.3, 4.0, 10.0): max |G_k - closed form| over k ∈ {0,π}, n = -6...5, both spins = 3.2e-15
(t, U, β) = (1.0, 0.5, 1.0): max |G_k - closed form| over k ∈ {0,π}, n = -6...5, both spins = 1.8e-16

limits of the closed form, each derivable by hand:

U = 0: G₀(0.3i) = 1.095890410959 - 0.41095890411im, 1/(0.3i + t) = 1.095890410959 - 0.41095890411im
t = 0: G₀(0.7i) = 0.0 - 0.321100917431im, atom $\frac{1}{2}\Sigma\pm(iv \pm U/2)^{-1} = 0.0 - 0.321100917431im$
T → 0: G₀(0.5i) = 0.595007770624 - 0.426591392052im, α/(iv-(t-c)) + α/(iv-(t+c)) = 0.595007770624 - 0.426591392052im
sum rule: z G₀(z) at z = 10⁷i → 1.000000000

trap 1, hopping sign flipped (t → -t):

spectrum unchanged: true
|G₀(-t) - G₀ closed form| = 0.532 <- D2 fails
|G₀(-t) - G_π closed form| = 6.7e-16 <- because G₀ and G_π were exchanged

D3 — local and nonlocal propagators

$G_{11} = \frac{1}{2}(G_0 + G_\pi)$ and $G_{12} = \frac{1}{2}(G_0 - G_\pi)$, both computed directly from the site operators $\mathcal{O} = (c_{i\sigma}, c_{j\sigma}^\dagger)$ rather than from G_k . By particle-hole symmetry $G_{11}(i\nu)$ is purely imaginary on the Matsubara axis; G_{12} comes out purely real.

In [5]:

```

for (t, U, β) in [(0.7, 2.3, 3.0), (0.3, 4.0, 10.0), (1.0, 0.5, 1.0)]
    ctx = DimerContext(HubbardDimerModel(t, U, β))
    e11 = e12 = r11 = 0.0
    for n in -4:3
        v = fermionic_freq(n)
        G0, Gπ = dimer_propagator(ctx, 0, v), dimer_propagator(ctx, 1, v)
        G11, G12 = dimer_site_propagator(ctx, 1, 1, v), dimer_site_propagator(ctx, 1, 2,
v)
        e11 = max(e11, abs(G11 - (G0 + Gπ)/2)); e12 = max(e12, abs(G12 - (G0 - Gπ)/2)); r
11 = max(r11, abs(real(G11)))
    end
    v = fermionic_freq(0)
    @printf("(%.1f, %.1f, %.4f): |G11 - ½(G0+Gπ)| = %.1e    |G12 - ½(G0-Gπ)| = %.1e
max|Re G11| = %.1e    G12(iv0) = %+.4f %+.1e\n",
        t, U, β, e11, e12, r11, real(dimer_site_propagator(ctx, 1, 2, v)), imag(dimer
_site_propagator(ctx, 1, 2, v)))
end

```

```

(0.7, 2.3, 3.0): |G11 - ½(G0+Gπ)| = 1.7e-16 |G12 - ½(G0-Gπ)| = 6.5e-17 max|Re G11|
= 8.1e-17 G12(iv0) = +0.2660 -6.1e-17i
(0.3, 4.0, 10.0): |G11 - ½(G0+Gπ)| = 5.8e-17 |G12 - ½(G0-Gπ)| = 5.0e-17 max|Re G11|
= 2.4e-15 G12(iv0) = +0.0509 -1.1e-16i
(1.0, 0.5, 1.0): |G11 - ½(G0+Gπ)| = 8.4e-17 |G12 - ½(G0-Gπ)| = 5.2e-17 max|Re G11|
= 4.9e-17 G12(iv0) = +0.0913 -5.0e-17i

```

D4 — Keldysh, $\ell = 2$

Per App. C, for $\ell = 2$ the prefactor of Eq. (67a) is 1 and $G^{[1]} = G^R$, $G^{[2]} = G^A$, $G^{[12]} = G^K$. The closed form continued as $i\nu \rightarrow \omega \pm i\gamma_0$ must reproduce G^R and G^A from Eq. (67). The fluctuation-dissipation theorem (C4), $G^K = (G^R - G^A) \tanh(\beta\omega/2)$ for fermions, is checked at each pole of each momentum component. At finite γ_0 it holds only up to $O(\gamma_0^2)$, as on the atom.

In [6]:

```

for (t, U, β) in [(0.7, 2.3, 3.0), (0.3, 4.0, 10.0), (1.0, 0.5, 1.0)]
    m = HubbardDimerModel(t, U, β); ctx = DimerContext(m); c = dimer_c(m)
    @printf("(t, U, β) = (%.1f, %.1f, %4.1f)\n", t, U, β)
    for q in 0:1
        A = momentum_op(ctx.ops, q, :up); Os = (A, permutedims(A)); cache = dimer_psf_cache(ctx, (:k, q, :up))
        G(ω, k, g) = keldysh_correlator(m, Os, [ω, -ω], k; γ0 = g, sp = ctx.sp, cache = cache)
        rel = maximum(max(abs(G(ω, [1], 0.05) - dimer_propagator_exact(m, q, ω + 0.05im)) / abs(G(ω, [1], 0.05))),
                        abs(G(ω, [2], 0.05) - dimer_propagator_exact(m, q, ω - 0.05im)) / abs(G(ω, [2], 0.05)))
        for ω in range(-3, 3; length = 13)
            @printf("    k = %s: G^R, G^A vs closed form at iv → ω ± iγ₀, 13 points: max rel. err %.1e\n", q == 0 ? "0" : "π", rel)
            poles = [t - c, t + c, U/2 - t, -(U/2 + t)] .* (q == 0 ? 1 : -1)
            for E in poles
                errs = [abs(G(E, [1,2], g) / (G(E, [1], g) - G(E, [2], g)) - tanh(β*E/2)) for g in (1e-2, 1e-3, 1e-4)]
                @printf("        FDT at pole ω = %+.3f: |G^K/(G^R-G^A) - tanh(βω/2)| = %.1e, %.1e, %.1e (γ₀ = 1e-2, 1e-3, 1e-4)\n", E, errs...)
            end
        end
    end
end

```

```

(t, U, β) = (0.7, 2.3, 3.0)
k = 0: G^R, G^A vs closed form at iv → ω ± iγ₀, 13 points: max rel. err 2.5e-14

```

FDT at pole $\omega = -1.112$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.6e-05, 1.6e-07, 1.6e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +2.512$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.6e-04, 1.6e-06, 1.6e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +0.450$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.5e-04, 2.5e-06, 2.5e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -1.850$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 7.4e-05, 7.4e-07, 7.4e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

$k = \pi$: G^R, G^A vs closed form at $iv \rightarrow \omega \pm i\gamma_0$, 13 points: max rel. err $7.6e-14$

FDT at pole $\omega = +1.112$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.6e-05, 1.6e-07, 1.6e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -2.512$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.6e-04, 1.6e-06, 1.6e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -0.450$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.5e-04, 2.5e-06, 2.5e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +1.850$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 7.4e-05, 7.4e-07, 7.4e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

$(t, U, \beta) = (0.3, 4.0, 10.0)$

$k = 0$: G^R, G^A vs closed form at $iv \rightarrow \omega \pm i\gamma_0$, 13 points: max rel. err $3.5e-14$

FDT at pole $\omega = -1.788$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.2e-05, 2.2e-07, 2.2e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +2.388$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 3.7e-05, 3.7e-07, 3.7e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +1.700$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 3.0e-05, 3.0e-07, 3.0e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -2.300$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.8e-05, 1.8e-07, 1.8e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

$k = \pi$: G^R, G^A vs closed form at $iv \rightarrow \omega \pm i\gamma_0$, 13 points: max rel. err $6.2e-14$

FDT at pole $\omega = +1.788$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.2e-05, 2.2e-07, 2.2e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -2.388$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 3.7e-05, 3.7e-07, 3.7e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -1.700$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 3.0e-05, 3.0e-07, 3.0e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +2.300$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 1.8e-05, 1.8e-07, 1.8e-09$ ($\gamma = 1e-2, 1e-3, 1e-4$)

$(t, U, \beta) = (1.0, 0.5, 1.0)$

$k = 0$: G^R, G^A vs closed form at $iv \rightarrow \omega \pm i\gamma_0$, 13 points: max rel. err $2.6e-14$

FDT at pole $\omega = -1.016$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 9.3e-07, 9.8e-09, 9.8e-11$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +3.016$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 5.3e-03, 5.3e-05, 5.3e-07$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -0.750$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.9e-04, 2.9e-06, 2.9e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -1.250$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.9e-04, 3.0e-06, 3.0e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

$k = \pi$: G^R, G^A vs closed form at $iv \rightarrow \omega \pm i\gamma_0$, 13 points: max rel. err $3.3e-14$

FDT at pole $\omega = +1.016$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 9.3e-07, 9.8e-09, 9.8e-11$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = -3.016$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 5.3e-03, 5.3e-05, 5.3e-07$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +0.750$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.9e-04, 2.9e-06, 2.9e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

FDT at pole $\omega = +1.250$: $|G^K/(G^R-G^A) - \tanh(\beta\omega/2)| = 2.9e-04, 3.0e-06, 3.0e-08$ ($\gamma = 1e-2, 1e-3, 1e-4$)

Four-point objects — what changes

Disconnected part. Every Wick contraction $\langle c_{k_a} c_{k_b}^\dagger \rangle$ is diagonal in momentum, so Eq. (73) (p. 15) gets a Kronecker delta per pairing. For

$$\mathcal{O} = (c_{k_1\sigma}, c_{k_2\sigma}^\dagger, c_{k_3\sigma'}, c_{k_4\sigma'}^\dagger),$$

$$G^{\text{dis}} = \beta G_{k_1}(i\omega_1) G_{k_3}(i\omega_3) (\delta_{\sigma\sigma'} \delta_{k_1 k_4} \delta_{k_2 k_3} \delta_{\omega_{23}, 0} - \delta_{k_1 k_2} \delta_{k_3 k_4} \delta_{\omega_{12}, 0}).$$

It is cross-checked against the kernel-free PSF split of Eq. (31), which needs no momentum bookkeeping at all.

Amputation in momentum space. In the site basis the external legs are 2×2 matrices G_{ij} , so the scalar division of Eq. (74) is wrong for $t \neq 0$ (trap 2, below). In the k basis the propagator is diagonal and Eq. (74) applies leg by leg:

$$F_{k_1 k_2 k_3 k_4}(i\omega) = \frac{G_{k_1 k_2 k_3 k_4}^{\text{con}}(i\omega)}{G_{k_1}(i\omega_1) G_{k_2}(-i\omega_2) G_{k_3}(i\omega_3) G_{k_4}(-i\omega_4)}, \quad \omega_4 = -\omega_{123}.$$

Cost, and a matrix-element filter. A 4p PSF runs over $16^4 \times 24 \approx 1.6 \times 10^6$ eigenstate cycles. H is not diagonal in the Fock basis here, so rotated matrix elements that a symmetry forbids come out as roundoff, $\sim 10^{-17}$, rather than exact zeros. Kept as they are, 99.9% of the cycles become "nonzero" terms with weights $\sim 10^{-33}$. `DimerContext` therefore zeroes rotated matrix elements below `otol = 1e-12` before multiplying. A genuine matrix element here is $\gtrsim 10^{-3}$ even at $U = 5 \times 10^{-3}$ ($\sqrt{\alpha_-} \approx U/8t$), and `otol = 0` gives the unfiltered sum for comparison.

In [7]:

```
m = HubbardDimerModel(0.7, 2.3, 3.0)
ctx = DimerContext(m)
ctx0 = DimerContext(m; otol = 0.0)
key = (:four, (0,0,0,0), :updn, :full)
n1 = sum(length(d.terms) for d in dimer_psf_cache(ctx, key))
n0 = sum(length(d.terms) for d in dimer_psf_cache(ctx0, key))
@printf("PSF terms of (c0↑, c1↑↑, c0↓, c1↓↓), summed over 24 permutations: %d unfiltered
d, %d with otol = 1e-12\n", n0, n1)

M(v...) = MatsubaraFreq.(collect(v))
FREQS = [M(1,3,5,-9), M(1,-1,3,-3), M(1,3,-1,-3), M(1,3,-3,-1), M(1,-1,1,-1), M(3,-1,-3,
1)]
FLABEL = ["generic", "ω12=0", "ω13=0", "ω14=0", "ω12=ω14=0", "ω13=0 (b)"]
@printf("filtered vs unfiltered G, 6 frequency points: max |diff| = %.1e\n",
maximum(abs(dimer_correlator_4p(ctx, (0,0,0,0), :updn, ms) - dimer_correlator_4p
(ctx0, (0,0,0,0), :updn, ms)) for ms in FREQS))
empty!(ctx0.cache)

e1 = e2 = 0.0
for qs in all_momentum_tuples(), s in (:updn, :upup), ms in FREQS
    global e1 = max(e1, abs(dimer_correlator_4p(ctx, qs, s, ms; part = :connected) - dimer_connected_4p(ctx, qs, s, ms)))
    global e2 = max(e2, abs(dimer_correlator_4p(ctx, qs, s, ms; part = :disconnected) - dimer_disconnected_4p(ctx, qs, s, ms)))
end
@printf("\nEq. (73) with momentum deltas vs the Eq. (31) PSF split, 16 tuples × 2 spins × 6 points:\n")
@printf("    max |G^con difference| = %.1e    max |G^dis difference| = %.1e\n", e1, e2)
```

```
PSF terms of (c0↑, c1↑↑, c0↓, c1↓↓), summed over 24 permutations: 397408 unfiltered, 272
with otol = 1e-12
filtered vs unfiltered G, 6 frequency points: max |diff| = 0.0e+00
Eq. (73) with momentum deltas vs the Eq. (31) PSF split, 16 tuples × 2 spins × 6 points:
    max |G^con difference| = 1.5e-15    max |G^dis difference| = 3.0e-15
```

D5 — momentum selection rule

Site exchange P maps $c_{0\sigma} \rightarrow c_{0\sigma}$ and $c_{\pi\sigma} \rightarrow -c_{\pi\sigma}$, and leaves H invariant. Every correlator with an odd number of $k = \pi$ operators is therefore odd

under P and vanishes identically: in every formalism and every Keldysh component. Equivalently, $F_{k_1 k_2 k_3 k_4} \neq 0$ only if $k_1 - k_2 + k_3 - k_4 \equiv 0$. This is the first test of how *non-identical* operators enter the permutation sum.

Two columns below: with the matrix-element filter, where forbidden tuples must be exactly zero, and without it, over all 1.6×10^6 cycles, where they must be zero to roundoff.

In [8]:

```
@printf(" q1q2q3q4  k1-k2+k3-k4  max|G| filtered (↑↑, ↑↑)  max|G| unfiltered (↑↑)\n")
for qs in all_momentum_tuples()
    f = [maximum(abs(dimer_correlator_4p(ctx, qs, s, ms)) for ms in FREQS) for s in (:updn, :upup)]
    u0 = maximum(abs(dimer_correlator_4p(ctx0, qs, :updn, ms)) for ms in FREQS)
    empty!(ctx0.cache) # 60 MB per unfiltered tuple: do
not keep
    @printf("  %d%d%d%d  %s  %.2e  %.2e  %.2e\n", qs..., momentum_al
lowed(qs) ? "allowed " : "FORBIDDEN", f..., u0)
end
```

q1q2q3q4	k1-k2+k3-k4	max G filtered (↑↑, ↑↑)		max G unfiltered (↑↑)
0000	allowed	1.06e+00	4.77e-01	1.06e+00
0001	FORBIDDEN	0.00e+00	0.00e+00	7.83e-16
0010	FORBIDDEN	0.00e+00	0.00e+00	6.74e-16
0011	allowed	1.51e+00	1.03e+00	1.51e+00
0100	FORBIDDEN	0.00e+00	0.00e+00	6.61e-16
0101	allowed	5.36e-01	5.61e-02	5.36e-01
0110	allowed	4.80e-01	1.03e+00	4.80e-01
0111	FORBIDDEN	0.00e+00	0.00e+00	5.17e-16
1000	FORBIDDEN	0.00e+00	0.00e+00	6.76e-16
1001	allowed	4.80e-01	1.03e+00	4.80e-01
1010	allowed	5.36e-01	5.61e-02	5.36e-01
1011	FORBIDDEN	0.00e+00	0.00e+00	4.19e-16
1100	allowed	1.51e+00	1.03e+00	1.51e+00
1101	FORBIDDEN	0.00e+00	0.00e+00	3.61e-16
1110	FORBIDDEN	0.00e+00	0.00e+00	3.93e-16
1111	allowed	1.06e+00	4.77e-01	1.06e+00

D6 — the $U = 0$ null test

At $U = 0$ Wick's theorem holds: $G^{\text{con}} = 0$ and $F = 0$ for all 16 k tuples and both spins. The spectrum is highly degenerate there, so the anomalous branch of Eq. (46) fires on many terms, both in S and in S^{dis} , and must cancel between them. Checked both ways: $G - G^{\text{dis}}$ from Eq. (73), and the connected PSF of Eq. (31).

In [9]:

```
for β in (3.0, 10.0)
    c0 = DimerContext(HubbardDimerModel(0.7, 0.0, β))
    a = maximum(abs(dimer_connected_4p(c0, qs, s, ms)) for qs in all_momentum_tuples(), s
in (:updn, :upup), ms in FREQS)
    b = maximum(abs(dimer_correlator_4p(c0, qs, s, ms; part = :connected)) for qs in all_
momentum_tuples(), s in (:updn, :upup), ms in FREQS)
    f = maximum(abs(dimer_vertex(c0, qs, s, ms)) for qs in all_momentum_tuples(), s in (:
updn, :upup), ms in FREQS)
    g = maximum(abs(dimer_correlator_4p(c0, qs, s, ms)) for qs in all_momentum_tuples(),
s in (:updn, :upup), ms in FREQS)
    @printf("U = 0, t = 0.7, β = %4.1f:  max|G - G^dis| = %.1e  max|G[S^con]| = %.1e  m
```

```
ax|F| = %.1e (max|G| = %.2f)\n", β, a, b, f, g)
end
```

U = 0, t = 0.7, β = 3.0: max|G - G^{dis}| = 1.6e-15 max|G[S^{con}]| = 1.0e-15 max|F| = 1.6e-14 (max|G| = 1.89)
 U = 0, t = 0.7, β = 10.0: max|G - G^{dis}| = 2.5e-14 max|G[S^{con}]| = 9.1e-15 max|F| = 8.7e-15 (max|G| = 16.99)

D7 — first order in U

p. 16, below Eq. (84): "the first term $U\delta_{\sigma,\bar{\sigma}'}$ on the right is the MF bare vertex $F_{0;\sigma\sigma'}$." Transforming the local $U\delta_{i_1=i_2=i_3=i_4}$ to the k basis,

$$F_{k_1 k_2 k_3 k_4; \uparrow \downarrow}^{(1)} = U \sum_{j=1,2} \frac{1}{4} e^{i(k_1 - k_2 + k_3 - k_4)(j-1)} = \frac{U}{2} \delta_{k_1 - k_2 + k_3 - k_4 \equiv 0}.$$

F/U is evaluated at $U = 10^{-2}$ and 5×10^{-3} and extrapolated linearly (Richardson) to $U \rightarrow 0$. The target is $\frac{1}{2}$ for every allowed tuple and every frequency, 0 for the forbidden ones, and 0 for $\uparrow\uparrow$ (no same-spin bare vertex).

In [10]:

```
Us = (1e-2, 5e-3)
cs = [DimerContext(HubbardDimerModel(0.7, U, 3.0)) for U in Us]
lim(qs, s, ms) = (v = [dimer_vertex(c, qs, s, ms) / U for (c, U) in zip(cs, Us)]; 2v[2] - v[1])
@printf(" q1q2q3q4 target lim F↑↑/U (max|dev| over 6 points) lim F↑↑/U (max|dev|)\n")
for qs in all_momentum_tuples()
    target = momentum_allowed(qs) ? 0.5 : 0.0
    L = [lim(qs, :updn, ms) for ms in FREQS]
    Luu = [lim(qs, :upup, ms) for ms in FREQS]
    @printf(" %d%d%d%d %.1f %+.6f (%.1e) %.1e\n", qs...,
    target, real(L[1]),
    maximum(abs, L .- target), maximum(abs, Luu))
end
```

q1q2q3q4	target	lim F↑↑/U (max dev over 6 points)	lim F↑↑/U (max dev)
0000	0.5	+0.499999 (5.2e-06)	2.4e-08
0001	0.0	+0.000000 (0.0e+00)	0.0e+00
0010	0.0	+0.000000 (0.0e+00)	0.0e+00
0011	0.5	+0.499999 (6.0e-06)	3.9e-08
0100	0.0	+0.000000 (0.0e+00)	0.0e+00
0101	0.5	+0.499999 (6.1e-06)	2.5e-08
0110	0.5	+0.499999 (6.0e-06)	3.9e-08
0111	0.0	+0.000000 (0.0e+00)	0.0e+00
1000	0.0	+0.000000 (0.0e+00)	0.0e+00
1001	0.5	+0.499999 (6.0e-06)	3.9e-08
1010	0.5	+0.499999 (6.1e-06)	2.5e-08
1011	0.0	+0.000000 (0.0e+00)	0.0e+00
1100	0.5	+0.499999 (6.0e-06)	3.9e-08
1101	0.0	+0.000000 (0.0e+00)	0.0e+00
1110	0.0	+0.000000 (0.0e+00)	0.0e+00
1111	0.5	+0.499999 (5.2e-06)	2.4e-08

D8 — $t \rightarrow 0$: the two branches of Eq. (46) against each other

At $t = 0$ the sites decouple, the site-basis vertex is $F^{\text{atom}} \delta_{i_1=i_2=i_3=i_4}$ with F^{atom} from Eq. (85), and the transformation of D7 gives

$$F_{k_1 k_2 k_3 k_4} \rightarrow \frac{1}{2} F^{\text{atom}} \delta_{k_1 - k_2 + k_3 - k_4 \equiv 0}.$$

At $t = 0$ exactly, the $N = 1$ bonding and antibonding levels are degenerate and the anomalous branch of Eq. (46) fires. At small $t \neq 0$ the same contribution comes from the regular branch instead, through denominators of order $2t$. Continuity across $t \rightarrow 0$ therefore tests the two branches against each other, which the atom alone could not do. All 16 tuples, both spins, and the six frequency points, including the δ -carrying ones.

In [11]:

```

U, β = 2.3, 3.0
ma = HubbardAtomModel(U, β)
# PSF terms that take the anomalous branch of Eq. (46), summed over the three single-δ po
ints and both
# spins. Terms with |w| < 1e-8 are skipped: see the note below the table.
function n_anomalous(c, qs)
    n = 0
    for ms in FREQS[2:4], s in (:updn, :upup), d in dimer_psf_cache(c, (:four, qs, s, :fu
ll)), t in d.terms
        abs(t.weight) > 1e-8 || continue
        _, van = composites([ms[d.p[i]] for i in 1:4], t.position, c.m.β)
        n += any(van)
    end
    n
end
dev(c, qsset) = maximum(abs(dimer_vertex(c, qs, s, ms) - (momentum_allowed(qs) ? 0.5 * ve
rtex_exact(ma, s, ms) : 0.0))
    for qs in qsset, s in (:updn, :upup), ms in FREQS)
@printf("    t      max |F - ½F^atom δ|    dev/t² (all)    dev/t² (0101)    dev/t² (0000)
anomalous terms: 0000    0101\n")
for t in (0.0, 1e-1, 1e-2, 1e-3, 3e-4, 1e-4, 3e-5, 1e-5)
    c = DimerContext(HubbardDimerModel(t, U, β))
    e = dev(c, all_momentum_tuples())
    if t == 0
        @printf("    0      %.3e
%3d    %3d\n", e,
            n_anomalous(c, (0,0,0,0)), n_anomalous(c, (0,1,0,1)))
    else
        @printf("    %.0e    %.3e          %8.3f          %10.3f          %9.3f
%3d    %3d\n", t, e, e/t^2,
            dev(c, [(0,1,0,1)]/t^2, dev(c, [(0,0,0,0)]/t^2, n_anomalous(c, (0,0,0,
0)), n_anomalous(c, (0,1,0,1)))
        end
    end
end
@printf("\nscale: max |½F^atom| = %.1f\n", maximum(abs(0.5 * vertex_exact(ma, s, ms)) for
s in (:updn, :upup), ms in FREQS))

```

t	max F - ½F ^{atom} δ	dev/t ² (all)	dev/t ² (0101)	dev/t ² (0000)	anomalous t
0	3.176e-14				432 384
1e-01	8.267e-01	82.665	52.159	82.665	
300 0					
1e-02	8.395e-03	83.953	52.915	83.953	
300 0					
1e-03	8.397e-05	83.966	52.923	83.966	
300 0					
3e-04	7.557e-06	83.966	52.848	83.966	
300 0					
1e-04	8.397e-07	83.966	42.738	83.966	
300 0					
3e-05	1.194e-06	1327.110	1327.110	83.966	
300 0					

1e-05 1.080e-05 107972.345 104766.627 205.159
300 0

scale: max $|\frac{1}{2}F^{\text{atom}}| = 21.1$

At $t = 0$ the dimer reproduces $\frac{1}{2}F^{\text{atom}}$ to machine precision. For $t \neq 0$ the deviation is a clean t^2 law (dev/ t^2 constant to five digits) from $t = 10^{-1}$ down to 10^{-3} , as it must be for a continuous limit.

Which terms switch branch depends on the tuple. The site exchange P is conserved, and the bonding and antibonding $N = 1$ levels, degenerate at $t = 0$, have opposite P . A composite frequency pairs two states with equal P when its two operators carry equal momenta, and opposite P otherwise.

- For 0000 the bonding-antibonding degeneracy never enters. The anomalous terms that disappear at $t \neq 0$ are those pairing the singlet with the triplet, degenerate at $t = 0$ and split by $c - U/2 \approx 4t^2/U$ after.
- For 0101 and 1010 the whole $\beta\delta_\omega$ contribution is anomalous at $t = 0$: 384 terms. At any $t \neq 0$ not one term is; the same contribution comes entirely from the regular branch through denominators $\pm 2t$.

The vertex is continuous across that switch: the cross-check of the two branches of Eq. (46) the atom could not provide.

Loss of precision, as the task expected. Below $t \approx 3 \times 10^{-4}$ the 0101 column departs from t^2 . Its numerical error grows as $1/t^2$: about 10^{-7} at $t = 10^{-4}$ and 10^{-5} at $t = 10^{-5}$. That is what the regular branch costs when two denominators are $\sim 2t$ and the terms must cancel to leave something finite. It crosses the physical t^2 deviation near $t \approx 2 \times 10^{-4}$. At $t = 10^{-5}$, 0000 shows a smaller error too. There the likely source is the eigenvectors: the singlet-triplet gap $4t^2/U \approx 2 \times 10^{-10}$ lets LAPACK mix the two states at $\varepsilon/\text{gap} \sim 10^{-6}$. That mixing is also why the PSF grows from 272 to several hundred terms at small t : symmetry-forbidden matrix elements of that size pass the 1e-12 filter. The term count above ignores weights below 10^{-8} for that reason. At $t = 0$ exactly none of this arises, and the anomalous branch gives the answer at machine precision. The practical rule: for a near-degeneracy, either resolve it exactly or stay well above $\sqrt[4]{\varepsilon}$ in the splitting parameter.

D9 — crossing and SU(2)

$F_{\uparrow\uparrow}(i\omega) = F_{\uparrow\downarrow}(i\omega) - F_{\uparrow\downarrow}(i\omega')$ (p. 16, below Eq. 85b), where ω' exchanges $\omega_1 \leftrightarrow \omega_3$, or equivalently $\omega_2 \leftrightarrow \omega_4$. The momentum labels are exchanged together with the frequencies. Both versions, all 16 tuples, six points.

In [12]:

```
e13 = e24 = 0.0; scale = 0.0
for qs in all_momentum_tuples(), ms in FREQS
    Fuu = dimer_vertex(ctx, qs, :upup, ms); Fud = dimer_vertex(ctx, qs, :updn, ms)
```

```

F13 = dimer_vertex(ctx, (qs[3], qs[2], qs[1], qs[4]), :updn, [ms[3], ms[2], ms[1], ms
[4]])
F24 = dimer_vertex(ctx, (qs[1], qs[4], qs[3], qs[2]), :updn, [ms[1], ms[4], ms[3], ms
[2]])
global e13 = max(e13, abs(Fuu - (Fud - F13))); global e24 = max(e24, abs(Fuu - (Fud -
F24)))
global scale = max(scale, abs(Fuu))
end
@printf("ω1↔ω3, k1↔k3: max |F↑↑ - (F↑↓ - F↑↓')| = %.1e\n", e13)
@printf("ω2↔ω4, k2↔k4: max |F↑↑ - (F↑↓ - F↑↓')| = %.1e      (max |F↑↑| = %.2f)\n", e24,
scale)

```

$\omega_{1\leftrightarrow\omega_3}, k_{1\leftrightarrow k_3}: \max |F_{\uparrow\uparrow} - (F_{\uparrow\downarrow} - F_{\uparrow\downarrow'})| = 3.6\text{e-}14$
 $\omega_{2\leftrightarrow\omega_4}, k_{2\leftrightarrow k_4}: \max |F_{\uparrow\uparrow} - (F_{\uparrow\downarrow} - F_{\uparrow\downarrow'})| = 4.3\text{e-}14 \quad (\max |F_{\uparrow\uparrow}| = 3.00)$

D10 — Keldysh

The same cached PSFs, with the kernel of Eq. (67b) and the prefactor

$2^{1-\ell/2} = \frac{1}{2}$ of Eq. (67a), at a real tuple ω with $\sum \omega_i = 0$ and $\gamma_0 = 0.1$:

- **D5 in the KF:** all 16 Keldysh components of the 8 forbidden tuples vanish.
- **D6 in the KF:** at $U = 0$, all 16 components of G^{con} vanish for all tuples and both spins.
- **Eq. (69)** for the fully retarded components:
 $2^{\ell/2-1} G^{[\eta]}(\omega) = \tilde{G}(i\omega \rightarrow \omega^{[\eta]})$, with $\tilde{G}$ the regular-kernel MF object of Eq. (68b), as on the atom.

The Keldysh vertex (amputation, Eq. 76) is not formed. It needs the Appendix D shifts, and D6's $F = 0$ follows from $G^{\text{con}} = 0$ anyway.

In [13]:

```

ω = [0.37, -1.1, 0.52, 0.21]; ω[4] = -sum(ω[1:3]); γ0 = 0.1
c0 = DimerContext(HubbardDimerModel(0.7, 0.0, 3.0))
fa = ff = con0 = 0.0
for qs in all_momentum_tuples(), s in (:updn, :upup), k in all_keldysh_components(4)
    v = abs(dimer_keldysh_4p(ctx, qs, s, ω, k; γ0 = γ0))
    momentum_allowed(qs) ? (global fa = max(fa, v)) : (global ff = max(ff, v))
    global con0 = max(con0, abs(dimer_keldysh_4p(c0, qs, s, ω, k; γ0 = γ0, part = :connec
ted)))
end
@printf("D5 (KF): max |G^k| allowed tuples = %.2f, forbidden tuples = %.1e (16 compo
nents × 2 spins)\n", fa, ff)
u0 = maximum(abs(dimer_keldysh_4p(ctx0, (1,0,0,0), :updn, ω, k; γ0 = γ0)) for k in all_ke
ldysh_components(4))
empty!(ctx0.cache)
@printf("      unfiltered PSFs, tuple 1000: max |G^k| = %.1e\n", u0)
@printf("D6 (KF): U = 0, max |G^con,k| over 16 tuples × 2 spins × 16 components = %.1e
\n", con0)
e = 0.0
for qs in all_momentum_tuples(), s in (:updn, :upup), η in 1:4, g in (0.1, 0.01)
    Os = dimer_operators_4p(ctx.ops, qs, s)
    lhs = 2 * dimer_keldysh_4p(ctx, qs, s, ω, [η]; γ0 = g)
    rhs = regular_sum(ctx.sp, Os, omega_eta(ω, η, g); cache = dimer_psf_cache(ctx, (:fou
r, qs, s, :full)))
    global e = max(e, abs(lhs - rhs))
end
@printf("Eq. (69): max |2 G^[η] - G̃(ω^[η])| over 16 tuples × 2 spins × η = 1...4 × γ0 ∈ {0.
1, 0.01} = %.1e\n", e)

```

D5 (KF): $\max |G^k|$ allowed tuples = 75.85, forbidden tuples = 0.0e+00 (16 components $\times$ 2 spins)
unfiltered PSFs, tuple 1000: $\max |G^k| = 5.4e-14$
D6 (KF): $U = 0$, $\max |G^{\text{con},k}|$ over 16 tuples $\times$ 2 spins $\times$ 16 components = 4.5e-14
Eq. (69): $\max |2 G^{\eta} - \tilde{G}(\omega^{\eta})|$ over 16 tuples $\times$ 2 spins $\times \eta = 1\dots 4 \times \gamma_0 \in \{0.1, 0.01\} = 0.0e+00$

Eq. (69) holds to the last bit. That is expected rather than lucky: with the p^{-1} bookkeeping right, the single-term kernel $K^{[\eta]}$ and the regular sum evaluated at $\omega^{[\eta]}$ perform the same arithmetic, so what the check tests is the prefactor and which slot receives the $+i(\ell - 1)\gamma_0$. The forbidden components are exact zeros because the filter removes the roundoff matrix elements; unfiltered they are zero to 10^{-16} .

Traps 2 and 3

Trap 2, site-basis amputation. The site vertex is obtained correctly by transforming the four legs of the momentum-space F ,

$F_{i_1 i_2 i_3 i_4} = \sum_k \prod_a u_{k_a i_a} F_{k_1 k_2 k_3 k_4}$ with $u = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Dividing the site-basis G_{1111}^{con} by four local legs G_{11} instead is wrong for $t \neq 0$ and happens to be right at $t = 0$.

Trap 3, accidental degeneracies. At $t = U/2$ the triplet level $-U$ meets $-U/2 - t$, so the anomalous branch fires where it normally would not. With an explicit tolerance for detecting $\Omega_{\bar{1} \dots \bar{j}} = 0$ (atol = 1e-10 on the spectral part, an exact integer test on the Matsubara part), the vertex is continuous through that point.

In [14]:

```
u = [1 1; 1 -1] / sqrt(2)
function site_vertex_vs_naive(c, ms)
    Fk = Dict{qs => dimer_vertex(c, qs, :updn, ms) for qs in all_momentum_tuples()}
    Gc = Dict{qs => dimer_connected_4p(c, qs, :updn, ms) for qs in all_momentum_tuples()}
    tr(D) = sum(prod(u[qs[a] + 1, 1] for a in 1:4) * D[qs] for qs in all_momentum_tuples())
    G11(f) = dimer_site_propagator(c, 1, 1, f)
    naive = tr(Gc) / (G11(ms[1]) * G11(MatsubaraFreq(-ms[2].m)) * G11(ms[3]) * G11(MatsubaraFreq(-ms[4].m)))
    return tr(Fk), naive
end
for t in (0.7, 0.0)
    c = DimerContext(HubbardDimerModel(t, 2.3, 3.0))
    for (ms, lab) in zip(FREQS[1:2], FLABEL[1:2])
        F, naive = site_vertex_vs_naive(c, ms)
        @printf("t = %.1f, %-8s F1111 (k-basis, transformed) = %.6f %.6fi naive site
division = %.6f %.6fi rel. diff %.1e\n",
            t, lab, real(F), imag(F), real(naive), imag(naive), abs(F - naive) / abs(F))
    end
end
@printf("t = 0 check: F1111 = F^atom from Eq. (85a): |diff| = %.1e\n\n",
    abs(site_vertex_vs_naive(DimerContext(HubbardDimerModel(0.0, 2.3, 3.0)), FREQS[1])[1] -
        vertex_exact(HubbardAtomModel(2.3, 3.0), :updn, FREQS[1])))

println("trap 3: t = U/2 = 1, where -U = -U/2 - t")
for (ms, lab) in zip(FREQS[1:2], FLABEL[1:2])
    F(t) = dimer_vertex(DimerContext(HubbardDimerModel(t, 2.0, 3.0)), (0,0,0,0), :updn, m
```

```

s)
    @printf("   %-8s F(1) = %+.8f, F(1 ± 1e-6) mean = %+.8f, |diff| = %.1e\n", lab, rea
l(F(1.0)),
           real((F(1.0 + 1e-6) + F(1.0 - 1e-6)) / 2), abs(F(1.0) - (F(1.0 + 1e-6) + F(1.
0 - 1e-6)) / 2))
end

```

```

t = 0.7, generic F1111 (k-basis, transformed) = +2.939958 +0.000000i naive site division
= +3.102248 +0.000000i rel. diff 5.5e-02
t = 0.7, ω12=0 F1111 (k-basis, transformed) = +5.025648 +0.000000i naive site division
= +5.725971 +0.000000i rel. diff 1.4e-01
t = 0.0, generic F1111 (k-basis, transformed) = +3.566017 +0.000000i naive site division
= +3.566017 +0.000000i rel. diff 2.2e-15
t = 0.0, ω12=0 F1111 (k-basis, transformed) = +7.417259 +0.000000i naive site division
= +7.417259 -0.000000i rel. diff 2.4e-15
t = 0 check: F1111 = Fatom from Eq. (85a): |diff| = 1.7e-15

```

```

trap 3: t = U/2 = 1, where -U = -U/2 - t
generic F(1) = +1.04236394, F(1 ± 1e-6) mean = +1.04236394, |diff| = 3.2e-13
ω12=0 F(1) = +0.95552008, F(1 ± 1e-6) mean = +0.95552008, |diff| = 5.9e-13

```

D11 — observation: what the singlet does to the $\beta\delta_\omega$ terms

In Eq. (85) the atom's $\beta u^2 \delta_\omega$ terms reflect its free local moment. In the dimer the two moments bind into a singlet, with singlet-triplet gap

$c - U/2 \approx 4t^2/U$ at large U . The question the task leaves open: is the $\delta_{\omega_{12},0}$ contribution to $F_{0000;\uparrow\downarrow}$ cut off once $\beta(c - U/2) \gg 1$?

Defining the " δ contribution" without fitting. Eq. (45) (p. 9) writes the MF kernel as a regular product, whose divergences cancel between cyclically related permutations, plus the anomalous term

$$-\frac{\beta}{2} \sum_j \delta_{\Omega_{\bar{1} \dots \bar{j}, 0}} \prod_{i \neq j} \Omega_{\bar{1} \dots \bar{i}}^{-1}.$$

The $\delta_{\omega_{12},0}$ contribution is this term restricted to vanishing composites whose slots are $\{1, 2\}$ or $\{3, 4\}$, applied to the connected PSF of Eq. (31) and amputated like F (`anomalous_part` , `dimer_delta_part`). Before using it on the dimer, the definition is validated on the atom. Applied to the atom, it must return exactly the three δ terms of Eq. (85a), channel by channel.

In [15]:

```

U, β = 2.3, 3.0
ma = HubbardAtomModel(U, β); opsa = operators(ma); spa = spectrum(ma)
u, th = U/2, tanh(β*U/4)
Osa = (opsa.d_up, opsa.dag_up, opsa.d_dn, opsa.dag_dn)
cachea = permuted_psfes(spa, Osa; part = :connected)
println("atom, Eq. (45) anomalous term per channel vs the δ terms of Eq. (85a):")
for (ms, lab) in zip(FREQS, FLABEL)
    G(f) = propagator_exact(ma, f)
    legs = G(ms[1]) * G(MatsubaraFreq(-ms[2].m)) * G(ms[3]) * G(MatsubaraFreq(-ms[4].m))
    iw = [im * value(f, β) for f in ms]; fac = β * u^2 * prod(iw .+ u) / prod(iw)
    errs = [abs(anomalous_part(ma, Osa, ms; channel = ch, sp = spa, cache = cachea) / leg
s - (ms[1].m + ms[ch[2]].m == 0) * coef * fac)
            for (ch, coef) in ([1,2], th), ([1,3], th - 1), ([1,4], th + 1))]
    @printf("   %-11s |diff| δ12: %.1e   δ13: %.1e   δ14: %.1e   (|βu²|(|iw+u|)/|iw|) = %.2

```

```
f)\n", lab, errs..., abs(fac))
end
c = DimerContext(HubbardDimerModel(0.0, U, β)); ms = FREQS[2]
iw = [im * value(f, β) for f in ms]
@printf("\ndimer at t = 0, tuple 0000: δ12 part / (½ × atom's) = %.12f\n",
        real(dimer_delta_part(c, (0,0,0,0), :updn, ms) / (0.5 * β * u^2 * th * prod(iw .+
u) / prod(iw))))
```

atom, Eq. (45) anomalous term per channel vs the δ terms of Eq. (85a):

generic	diff	δ_{12} : 0.0e+00	δ_{13} : 0.0e+00	δ_{14} : 0.0e+00	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 6.4$
7)					
$\omega_{12}=0$	diff	δ_{12} : 3.7e-15	δ_{13} : 0.0e+00	δ_{14} : 0.0e+00	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 9.9$
2)					
$\omega_{13}=0$	diff	δ_{12} : 0.0e+00	δ_{13} : 3.5e-16	δ_{14} : 0.0e+00	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 9.9$
2)					
$\omega_{14}=0$	diff	δ_{12} : 0.0e+00	δ_{13} : 0.0e+00	δ_{14} : 1.5e-14	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 9.9$
2)					
$\omega_{12}=\omega_{14}=0$	diff	δ_{12} : 0.0e+00	δ_{13} : 0.0e+00	δ_{14} : 2.1e-14	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 19.$
31)					
$\omega_{13}=0$ (b)	diff	δ_{12} : 0.0e+00	δ_{13} : 6.7e-16	δ_{14} : 0.0e+00	$(\beta u^2 \prod(i\omega+u)/\prod i\omega = 9.9$
2)					

dimer at t = 0, tuple 0000: δ_{12} part / ($\frac{1}{2} \times$ atom's) = 1.000000000000

The scan. $U = 4$, three hoppings $t = 0.2, 0.4, 0.8$ (gaps $c - U/2 = 0.040, 0.154, 0.561$), and β swept so that $x = \beta(c - U/2)$ runs from 10^{-2} to 40. The frequencies are the lowest ones on the $\omega_{12} = 0$ line, $\omega = (\nu_0, -\nu_0, \nu_1, -\nu_1)$, so they move with β . ω_{13} and ω_{14} stay nonzero, so only the $\delta_{\omega_{12}}$ channel is active.

The raw size $|D_{12}|$ mixes two effects. The atom's term $\beta u^2 \text{th} \prod(i\omega + u) / \prod(i\omega)$ grows like β^5 at these frequencies, because $\prod i\omega \propto \beta^{-4}$. What the question is about is the singlet, so the plotted quantity is the ratio to the free-moment value at the same β and frequencies, $R = |D_{12}^{\text{dimer}}| / |\frac{1}{2} D_{12}^{\text{atom}}|$. $R = 1$ means "as if the moments were free". Points where D_{12} is below 10^{-13} of the canceling S and S^{dis} pieces are dropped as roundoff.

In [16]:

```
function delta_scan(t; U = 4.0, xs = 10.^range(-2, log10(40); length = 22))
    gap = dimer_c(HubbardDimerModel(t, U, 1.0)) - U/2
    rows = NamedTuple[]
    for x in xs
        β = x / gap
        c = DimerContext(HubbardDimerModel(t, U, β))
        ms = MatsubaraFreq.([1, -1, 3, -3])
        D = dimer_delta_part(c, (0,0,0,0), :updn, ms)
        # roundoff floor: the connected anomalous term is a difference of S and S^dis pie
ces
        Os = dimer_operators_4p(c.ops, (0,0,0,0), :updn)
        a(part) = abs(anomalous_part(c.m, Os, ms; channel = [1,2], sp = c.sp,
                                cache = dimer_psf_cache(c, (:four, (0,0,0,0), :updn,
part))))
        u = U/2; iw = [im * value(f, β) for f in ms]
        atom = 0.5 * β * u^2 * tanh(β*u/2) * prod(iw .+ u) / prod(iw)
        push!(rows, (x = x, β = β, D = abs(D), R = abs(D) / abs(atom), ok = a(:connected)
> 1e-13 * a(:full)))
    end
    return gap, rows
end
scans = Dict{t => delta_scan(t) for t in (0.2, 0.4, 0.8)}
for t in (0.2, 0.4, 0.8)
```

```

gap, rows = scans[t]
@printf("t = %.1f, gap c - U/2 = %.4f\n      x =  $\beta$ ·gap       $\beta$       |D12|      R
= |D12|/| $\frac{1}{2}$ D12atom|\n", t, gap)
  for r in rows[1:3:end]
    @printf("      %9.3e      %9.3f      %10.3e      %10.3e %s\n", r.x, r. $\beta$ , r.D, r.R, r.ok ? ""
: " (roundoff, dropped)")
  end
end
end

```

```

t = 0.2, gap c - U/2 = 0.0396
      x =  $\beta$ ·gap       $\beta$       |D12|      R = |D12|/| $\frac{1}{2}$ D12atom|
1.000e-02      0.252      1.282e-01      9.982e-01
3.270e-02      0.826      1.451e+00      9.849e-01
1.069e-01      2.700      2.615e+01      9.303e-01
3.497e-01      8.830      2.009e+03      7.737e-01
1.144e+00      28.876      7.069e+04      9.368e-02
3.740e+00      94.432      2.635e+03      9.586e-06
1.223e+01      308.816      1.318e+00      1.285e-11
4.000e+01      1009.902      4.527e-11      1.180e-24      (roundoff, dropped)

t = 0.4, gap c - U/2 = 0.1541
      x =  $\beta$ ·gap       $\beta$       |D12|      R = |D12|/| $\frac{1}{2}$ D12atom|
1.000e-02      0.065      8.426e-03      9.995e-01
3.270e-02      0.212      9.012e-02      9.949e-01
1.069e-01      0.694      9.727e-01      9.551e-01
3.497e-01      2.270      1.326e+01      7.843e-01
1.144e+00      7.424      2.743e+02      2.274e-01
3.740e+00      24.277      2.397e+01      7.473e-05
1.223e+01      79.391      9.666e-03      8.364e-11
4.000e+01      259.629      7.422e-13      1.723e-23      (roundoff, dropped)

t = 0.8, gap c - U/2 = 0.5612
      x =  $\beta$ ·gap       $\beta$       |D12|      R = |D12|/| $\frac{1}{2}$ D12atom|
1.000e-02      0.018      6.348e-04      9.999e-01
3.270e-02      0.058      6.782e-03      9.984e-01
1.069e-01      0.191      7.173e-02      9.836e-01
3.497e-01      0.623      6.982e-01      8.597e-01
1.144e+00      2.038      5.053e+00      4.027e-01
3.740e+00      6.664      1.744e+00      2.296e-03
1.223e+01      21.793      1.911e-03      1.012e-08
4.000e+01      71.270      2.776e-15      4.116e-23      (roundoff, dropped)

```

In [17]:

```

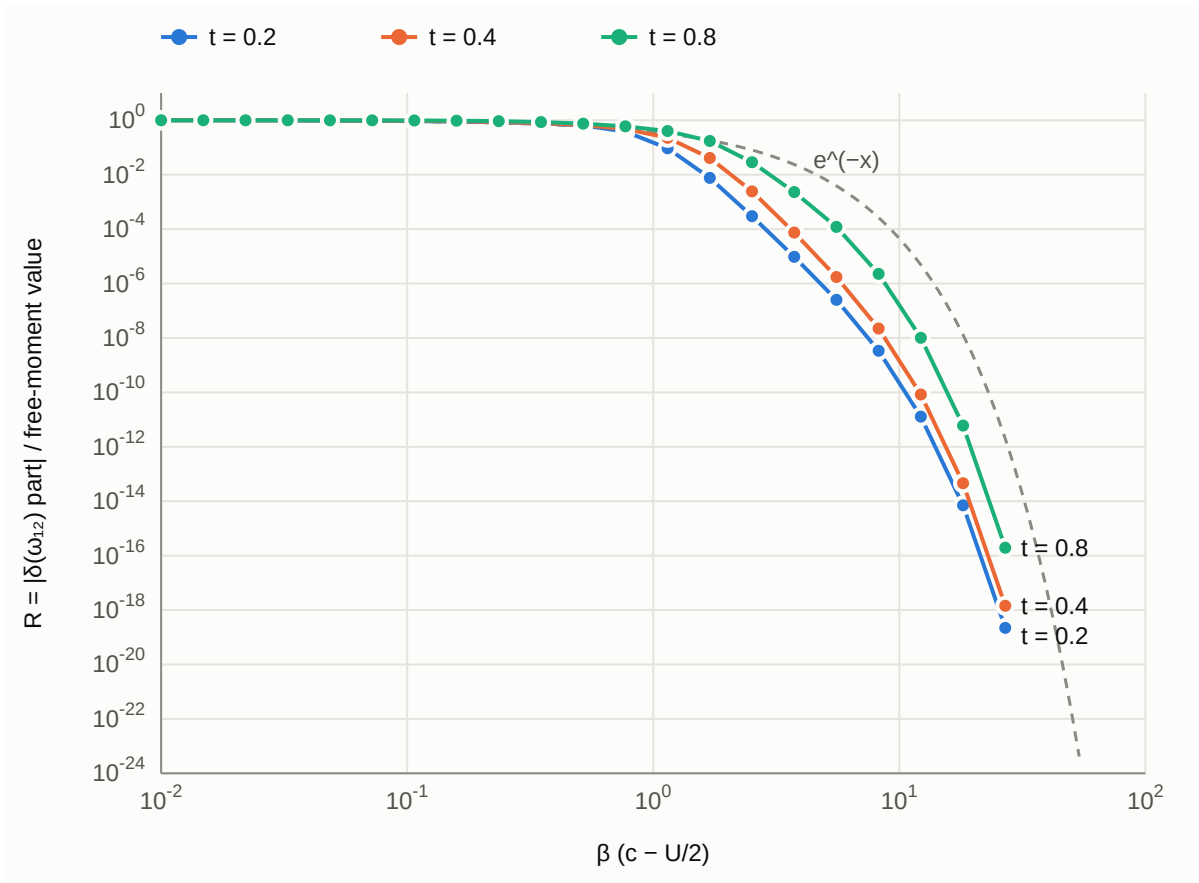
# Static SVG, stdlib only. Validated categorical slots 1-3 (all-pairs CVD-safe); direct l
abels at the
# line ends because slot 3 is below 3:1 contrast; hover a marker for its values.
function loglog_svg(series; xlab, ylab, xlim, ylim, guide = nothing)
  W, H, l, r, tp, b = 680, 440, 78, 110, 44, 56
  X(x) = l + (log10(x) - log10(xlim[1])) / (log10(xlim[2]) - log10(xlim[1])) * (W - l -
r)
  Y(y) = tp + (log10(ylim[2]) - log10(y)) / (log10(ylim[2]) - log10(ylim[1])) * (H - tp
- b)
  io = IOBuffer()
  print(io, <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 $W $H" width="$W" h
eight="$H" font-family="Helvetica, Arial, sans-serif" font-size="12" role="img" aria-labe
l="$ylab against $xlab">
<style>.bg{fill:#fcfcfb}.ink{fill:#0b0b0b}.ink2{fill:#52514e}.grid{stroke:#e4e3df;stroke-
width:1}.axis{stroke:#8a8983;stroke-width:1}.guide{stroke:#8a8983;stroke-width:1.5;stroke
-dasharray:5 4;fill:none}
@media (prefers-color-scheme: dark){.bg{fill:#1a1a19}.ink{fill:#ffffff}.ink2{fill:#c3c2b
7}.grid{stroke:#383835}.axis{stroke:#6f6e69}}</style>
<rect class="bg" x="0" y="0" width="$W" height="$H"/>""")
  for e in ceil(Int, log10(xlim[1])):floor(Int, log10(xlim[2]))
    x = X(10.0e)
    print(io, <line class="grid" x1="$x" y1="$tp" x2="$x" y2="$ (H-b)"/><text class
="ink2" x="$x" y="$ (H-b+18)" text-anchor="middle">10<tspace dy="-6" font-size="9">$e</tspace
n></text>""")
  end
  for e in ceil(Int, log10(ylim[1])):2:floor(Int, log10(ylim[2]))
    y = Y(10.0e)
    print(io, <line class="grid" x1="$l" y1="$y" x2="$ (W-r)" y2="$y"/><text class
="ink2" x="$ (l-8)" y="$ (y+4)" text-anchor="end">10<tspace dy="-6" font-size="9">$e</tspace
n></text>""")
  end
end

```

```

end
print(io, ""<line class="axis" x1="$l" y1="$ (H-b)" x2="$ (W-r)" y2="$ (H-b)"/><line cl
ass="axis" x1="$l" y1="$tp" x2="$l" y2="$ (H-b)"/>""")
print(io, ""<text class="ink" x="$ ((l+W-r)/2)" y="$ (H-12)" text-anchor="middle">$xla
b</text>""")
print(io, ""<text class="ink" transform="translate(18,$ ((tp+H-b)/2)) rotate(-90)" te
xt-anchor="middle">$ylab</text>""")
if guide != nothing
    f, lab = guide
    xs = 10 .^ range(log10(xlim[1]), log10(xlim[2]); length = 120)
    pts = join(["$(X(x)),$(Y(f(x)))" for x in xs if ylim[1] <= f(x) <= ylim[2]], " ")
    print(io, ""<polyline class="guide" points="$pts"/>""")
    xg = 4.0
    print(io, ""<text class="ink2" x="$ (X(xg)+6)" y="$ (Y(f(xg)))">$lab</text>""")
end
# direct labels at the line ends, pushed apart to at least 15 px
ends = sort([(Y(s[4][end]), i) for (i, s) in enumerate(series)])
laby = Dict{Int,Float64}()
for (k, (y, i)) in enumerate(ends)
    laby[i] = k == 1 ? y : max(y, laby[ends[k-1][2]] + 15)
end
for (i, (name, color, xs, ys)) in enumerate(series)
    pts = join(["$(X(x)),$(Y(y))" for (x, y) in zip(xs, ys)], " ")
    print(io, ""<polyline points="$pts" fill="none" stroke="$color" stroke-width="2"
stroke-linejoin="round"/>""")
    for (x, y) in zip(xs, ys)
        print(io, ""<circle cx="$ (X(x))" cy="$ (Y(y))" r="4" fill="$color" stroke="#f
cfcfb" stroke-width="2"><title>$name: x = $(round(x, sigdigits=3)), R = $(round(y, sigdig
its=3))</title></circle>""")
    end
    print(io, ""<text class="ink" x="$ (X(xs[end])+8)" y="$ (laby[i]+4)">$name</text
>""")
    # legend row above the plot
    lx = l + (i - 1) * 110
    print(io, ""<line x1="$lx" y1="16" x2="$ (lx+18)" y2="16" stroke="$color" stroke-
width="2"/><circle cx="$ (lx+9)" cy="16" r="4" fill="$color"/><text class="ink" x="$ (lx+2
4)" y="20">$name</text>""")
    end
    print(io, "</svg>")
    return String(take!(io))
end
colors = Dict{0.2 => "#2a78d6", 0.4 => "#eb6834", 0.8 => "#1baf7a"}
series = [("$t = $t", colors[t], [r.x for r in scans[t][2] if r.ok], [r.R for r in scans
[t][2] if r.ok]) for t in (0.2, 0.4, 0.8)]
display("image/svg+xml", loglog_svg(series; xlab = " $\beta (c - U/2)$ ", ylab = " $R = |\delta(\omega_{12})|$  par
t / free-moment value",
    xlim = (1e-2, 1e2), ylim = (1e-24, 10.0), guide = (x -> exp(-x), "e(-x)"))

```



In [18]:

```
# the decay rate: -d ln|D₁₂| / dβ at large x, against the singlet-triplet gap
println("  t      gap c-U/2      local rate -Δln|D₁₂|/Δβ  (last two scan points)      rate/ga
p  (gap - rate)·β")
for t in (0.2, 0.4, 0.8)
    gap, rows = scans[t]
    ok = [r for r in rows if r.ok]
    a, b = ok[end-1], ok[end]
    rate = -(log(b.D) - log(a.D)) / (b.β - a.β)
    @printf("  %.1f      %.4f      %.4f  (x = %.1f → %.1f)      %.3f
%.2f\n", t, gap, rate, a.x, b.x,
           rate / gap, (gap - rate) * sqrt(a.β * b.β))
end
```

t	gap c-U/2	local rate -Δln D₁₂ /Δβ (last two scan points)	rate/gap	(gap - rate)·β
0.2	0.0396	0.0378 (x = 18.2 → 26.9)	0.955	1.00
0.4	0.1541	0.1472 (x = 18.2 → 26.9)	0.955	0.99
0.8	0.5612	0.5345 (x = 18.2 → 26.9)	0.952	1.05

Reading D11. The expectation holds, and more sharply than stated.

- For $\beta(c - U/2) \ll 1$ the ratio $R \rightarrow 1$. The two sites behave as free moments, and the dimer's $\delta_{\omega_{12}}$ term is half the atom's, as the D8 transformation predicts.
- The crossover sits at $\beta(c - U/2) \approx 1$ for every t .
- Beyond it, $|D_{12}|$ itself decays exponentially at a rate set by the singlet-triplet gap. At $x \approx 20$ the local rate $-d \ln |D_{12}| / d\beta$ is 0.95 of $c - U/2$ for all three t . The shortfall times β is ≈ 1 in each case, so it is a power-law prefactor: $|D_{12}| \approx C \beta e^{-\beta(c - U/2)}$. $e^{-\beta(c - U/2)}$ is the Boltzmann weight of the thermally excited triplet relative to the singlet. Only the degenerate

triplet can carry a Curie-like $\beta\delta_\omega$ term. The nondegenerate singlet's own diagonal cycles are cancelled by S^{dis} , since a pure state factorises.

- R falls faster than e^{-x} (dashed guide). The free-moment reference it is divided by keeps growing like β^5 at these frequencies, so $R \sim e^{-x}/x^{\sim 4-5}$.

This is an observation from the numerics, not a derivation. It fits the physical picture: once the moments bind, the static spin susceptibility behind the δ term changes from Curie to activated.

Summary

Check	Result
operators	canonical anticommutators exact; missing strings caught; $[H, P] = 0$
D1	levels, degeneracies and Z to $\leq 5\text{e-}15$; shifted weights finite at $\beta = 1000$
D2	G_k vs closed form $\leq 4\text{e-}15$ at three parameter sets; the three limits; wrong hopping sign caught
D3	G_{11}, G_{12} from site operators $\leq 1\text{e-}15$; G_{11} purely imaginary
D4	$G^R, G^A \leq 1\text{e-}13$ relative; FDT (C4) at all 8 poles, $O(\gamma_0^2)$
D5	8 forbidden tuples exactly 0 ($\leq 8\text{e-}16$ unfiltered); allowed tuples finite
D6	$U = 0$: $G^{\text{con}}, F \leq 3\text{e-}14$
D7	$F_{\uparrow\downarrow}/U \rightarrow \frac{1}{2}\delta$, $F_{\uparrow\uparrow}/U \rightarrow 0$, to $6\text{e-}6$
D8	$t = 0$ exact; deviation $\propto t^2$ for $10^{-3} \leq t \leq 10^{-1}$; anomalous and regular branches agree; roundoff $\propto 1/t^2$ takes over below $t \approx 2 \times 10^{-4}$
D9	both crossing forms $\leq 5\text{e-}14$
D10	KF selection rule exact, $U = 0$ null $\leq 5\text{e-}14$, Eq. (69) exact
D11	$\delta_{\omega_{12}}$ term: free-moment value for $x \ll 1$, cut off as $e^{-\beta(c-U/2)}$ for $x \gg 1$